Developments in selective breeding

Success with wheat breeding

William Farrer was one of the first people to improve wheat crops so they could grow in Australian conditions. Farrer developed the basis for a good wheat crop, which has heads level for harvesting, high grain yield, makes excellent flour and resists disease – especially a mould called rust.

Problem with wheat breeding

The sets of genes that plants in a crop have are almost identical. This makes them susceptible to disease. The wild originals of the plant may have genes for disease resistance that are no longer present in the crop plants.

During the mid-1900s, Russian scientist Nikolai Vavilov and others, such as CSIRO's Sir Otto Frankel, suggested that wild seeds of crops be placed in freezer storage in case we ever need to reverse the specialisation of crops by **back-breeding**.

Word bank

resistant
high
bred
rust
term
seeds
crops
disease

☞ **1** Use the word bank to complete the following sentences.

Farrer _____ wheat that was _____ yield and _____ to _____.

Crops such as grain _____ are susceptible to _____ and we guard against long-_____ problems by keeping wild _____ in cold storage.

2 The table below lists developments that have occurred as a result of selective breeding. Write whether you think each development is a success or a problem.

Development	Success or problem?
Much higher yielding rice has been developed.	
Rice that grows with less water has been bred.	
Death at birth in some farm animals is increasing.	
Scientists can produce genetically modified food crops.	

3 Why might selective breeding lead to problems at birth? _____

_____

© Keith McTaggart and Blake Education 2012
© Blake Education 978 1 92571 038 0 Achieve! Science as a Human Endeavour Biology Yr 7 - 8